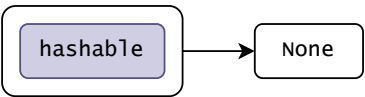


Основные методы множеств

add(element)

Добавление элемента

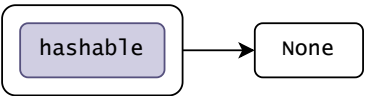
```
numbers = {1, 2, 3, 4, 5}
numbers.add(9)
print(numbers)  # {1, 2, 3, 4, 5, 9}
```



remove(element)

Удаление элемента

```
numbers = {1, 2, 3, 4, 5}
numbers.remove(1)
print(numbers)  # {2, 3, 4, 5}
numbers.remove(9)  # ! KeyError
```

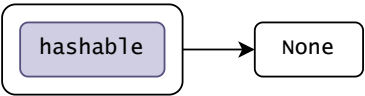


discard(element)

Удаление элемента

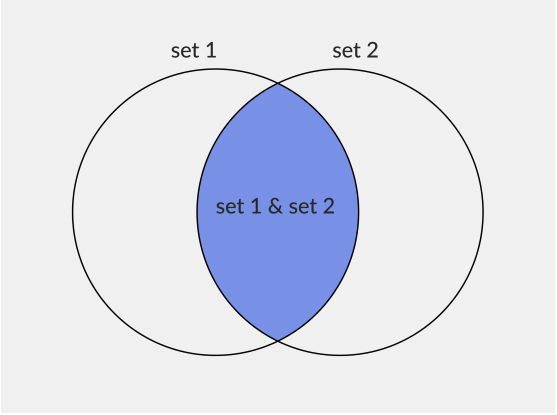
Если нет элемента,
ошибки не будет

```
numbers = {1, 2, 3, 4, 5}
numbers.discard(1)
print(numbers)  # {2, 3, 4, 5}
numbers.discard(9)
print(numbers)  # {2, 3, 4, 5}
```



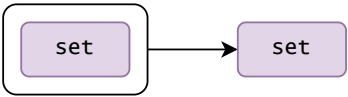
intersection(set)

Пересечение



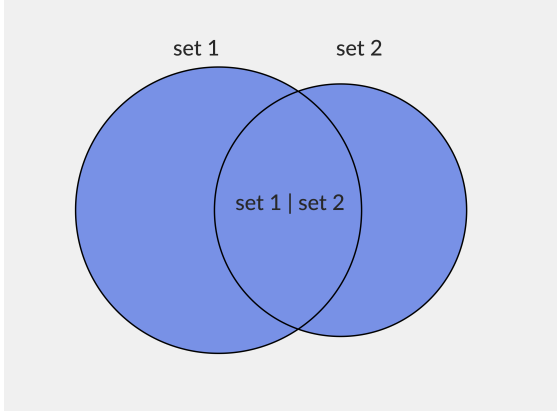
```
numbers1 = {1, 2, 3, 4, 5}
numbers2 = {4, 5, 6, 7, 8}

print(numbers1.intersection(numbers2))
print(numbers1 & numbers2) # {4, 5}
```



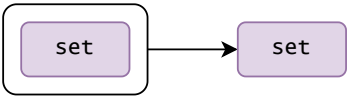
union(set)

Объединение



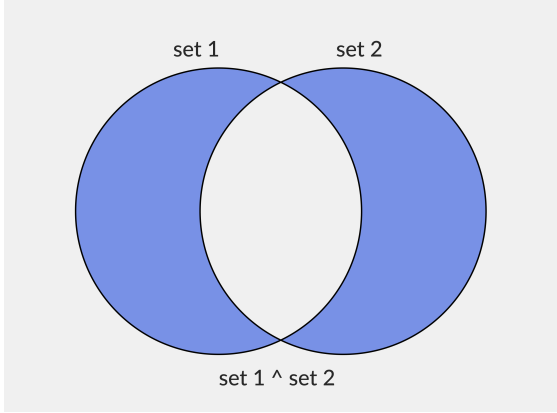
```
numbers1 = {1, 2, 3, 4, 5}
numbers2 = {4, 5, 6, 7, 8}

print(numbers1.union(numbers2))
print(numbers1 | numbers2) # {1, 2, 3, 4, 5, 6, 7, 8}
```



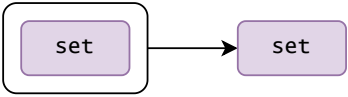
symmetric_difference(set)

Симметрическая разница



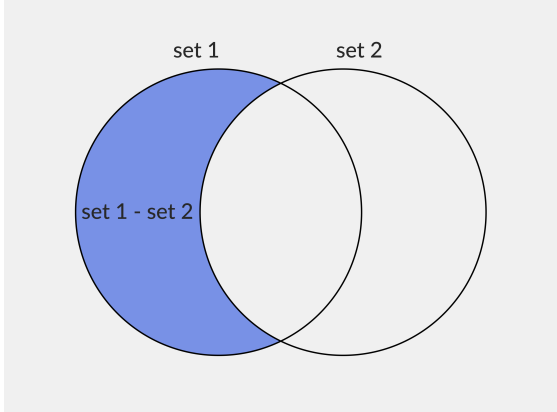
```
numbers1 = {1, 2, 3, 4, 5}
numbers2 = {4, 5, 6, 7, 8}

print(numbers1.symmetric_difference(numbers2))
print(numbers1 ^ numbers2) # {1, 2, 3, 6, 7, 8}
```



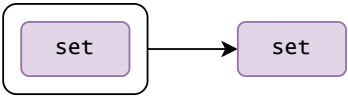
difference(set)

Разница

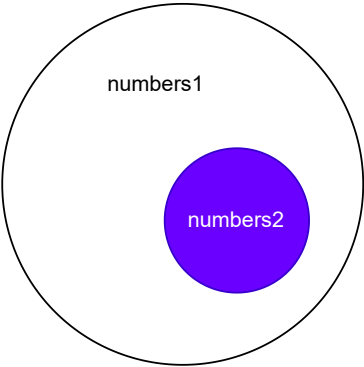


```
numbers1 = {1, 2, 3, 4, 5}
numbers2 = {4, 5, 6, 7, 8}

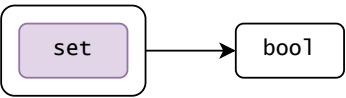
print(numbers1.difference(numbers2))
print(numbers1 - numbers2) # {1, 2, 3}
```



Является подмножеством другого



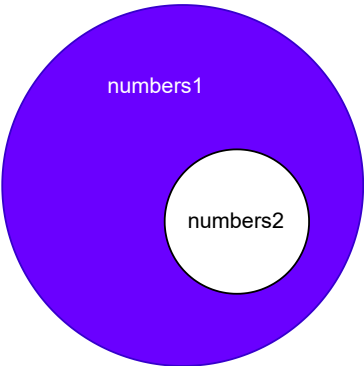
issubset(set)



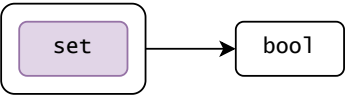
```
numbers1 = {1, 2, 3, 4, 5}
numbers2 = {2, 5}

print(numbers2.issubset(numbers1)) # True
```

Множество содержит другое



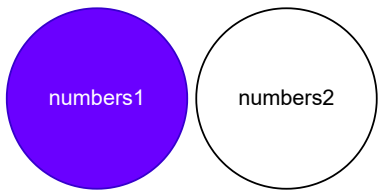
issuperset(set)



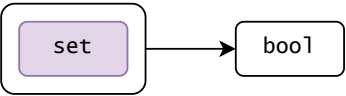
```
numbers1 = {1, 2, 3, 4, 5}
numbers2 = {2, 5}

print(numbers1.issuperset(numbers2)) # True
```

Множество не имеет пересечения с другим



issuperset(set)



```
numbers1 = {1, 2, 3, 4, 5}
numbers2 = {6, 7, 8, 9, 10}

print(numbers1.isdisjoint(numbers2)) # True
```